Object Detection

Use Python to demonstrate **Ultralytics**: Object Detection effects on images, videos, and real-time detection.

1. Model Introduction

Object detection is a task that involves identifying the location and class of objects in images or video streams.

The output of an object detector is a set of bounding boxes that surround objects in the image, along with class labels and confidence scores for each box. If you need to identify objects of interest in a scene but don't need to know the exact position or precise shape of the objects, then object detection is a good choice.

2. Target Prediction: Image

Use yolo26n.engine to predict images in the ultralytics26 folder.

Enter the code folder:

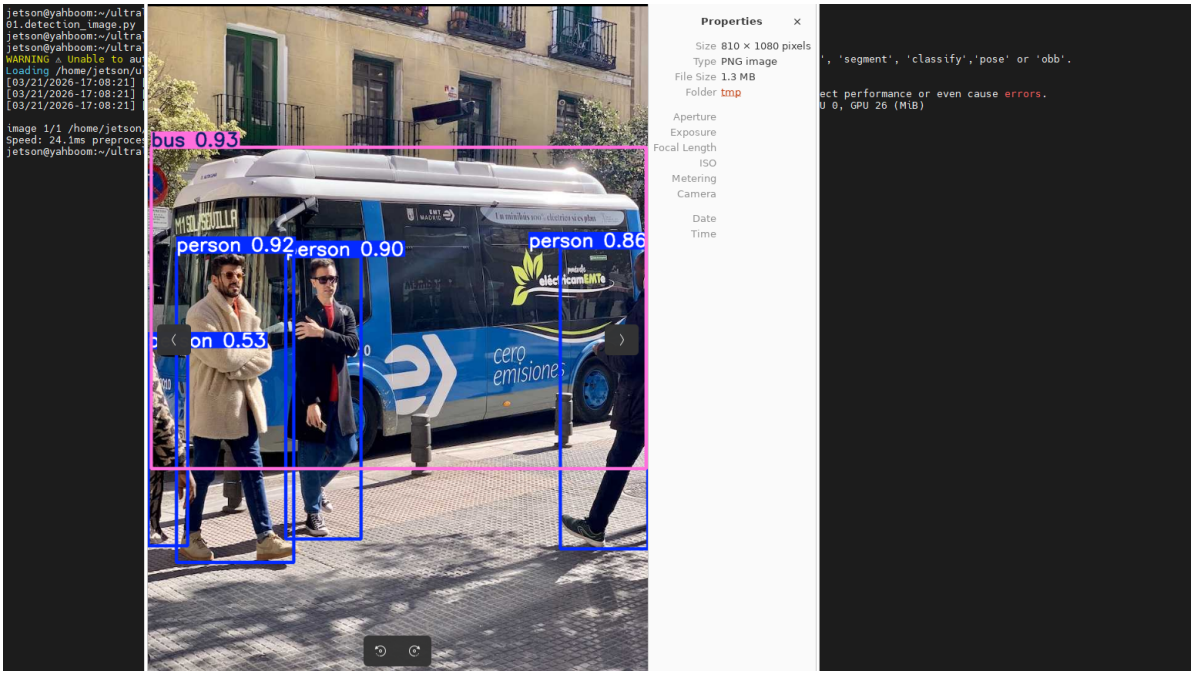
```
cd /home/jetson/ultralytics/yahboom_demo
```

Run the code:

```
python3 01.detection_image.py
```

Effect Preview

yolo recognition output image location: /home/jetson/ultralytics/output/



Example code:

```
from ultralytics import YOLO

# Load a model
model = YOLO("/home/jetson/ultralytics/yolo26n.engine")

# Run batched inference on a list of images
results = model("/home/jetson/ultralytics/assets/bus.jpg")  # return a list of
Results objects

# Process results list
for result in results:
    boxes = result.boxes  # Boxes object for bounding box outputs
    # masks = result.masks  # Masks object for segmentation masks outputs
    # keypoints = result.keypoints  # Keypoints object for pose outputs
    # probs = result.probs  # Probs object for classification outputs
    # obb = result.obb  # Oriented boxes object for OBB outputs
    result.show()  # display to screen
    result.save(filename="/home/jetson/ultralytics/output/bus_output.jpg")  #
save to disk
```

3. Target Prediction: Video

Use yolo26n_ncnn_model to predict videos in the ultralytics26 folder (not built-in ultralytics videos).

Enter the code folder:

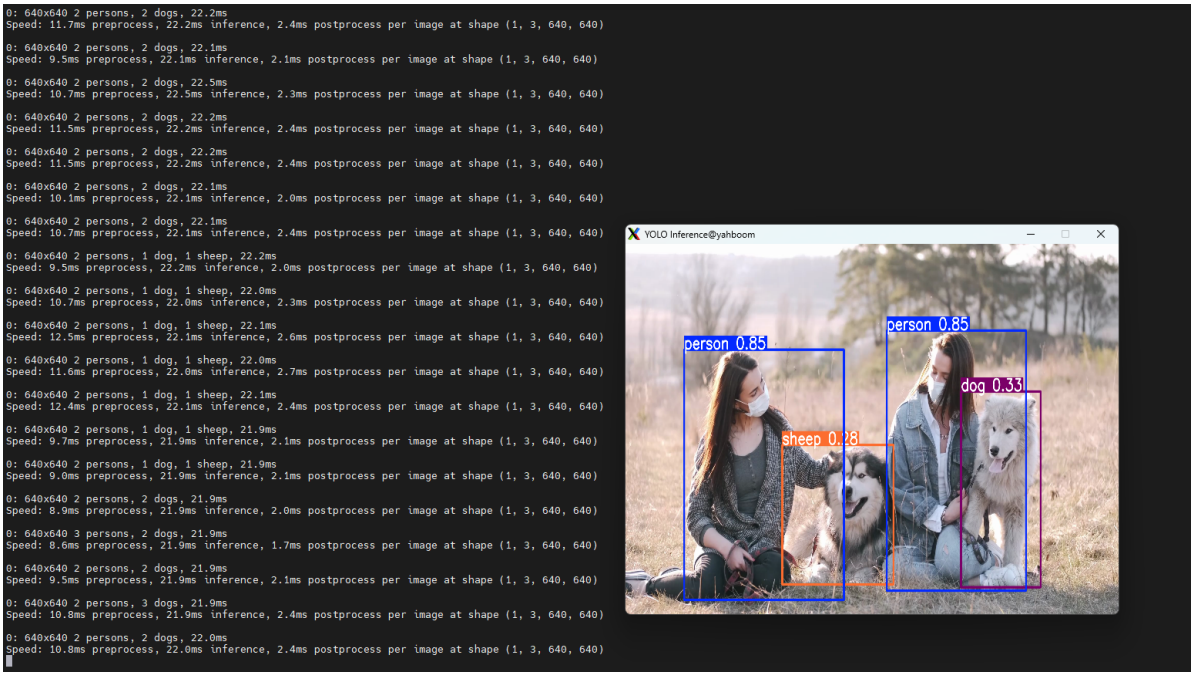
```
cd /home/jetson/ultralytics/yahboom_demo
```

Run the code:

```
python3 01.detection_video.py
```

Effect Preview

yolo recognition output video location: /home/jetson/ultralytics/output/



Example code:

```
import cv2
from ultralytics import YOLO

# Load the YOLO model
model = YOLO("/home/jetson/ultralytics/yolo26n.engine")

# Open the video file
video_path = "/home/jetson/ultralytics/ultralytics/videos/people_animals.mp4"
cap = cv2.VideoCapture(video_path)

# Get the video frame size and frame rate
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = int(cap.get(cv2.CAP_PROP_FPS))

# Define the codec and create a VideoWriter object to output the processed video
output_path = "/home/jetson/output/01.people_animals_output.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v')  # You can use 'XVID' or 'mp4v'
depending on your platform
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

# Loop through the video frames
while cap.isOpened():
    # Read a frame from the video
    success, frame = cap.read()

    if success:
        # Run YOLO inference on the frame
        results = model(frame)

        # Visualize the results on the frame
        annotated_frame = results[0].plot()
```

```
        # Write the annotated frame to the output video file
        out.write(annotated_frame)

        # Display the annotated frame
        cv2.imshow("YOLO Inference", cv2.resize(annotated_frame, (640, 480)))

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        # Break the loop if the end of the video is reached
        break
# Release the video capture and writer objects, and close the display window
cap.release()
out.release()
cv2.destroyAllWindows()
```

4. Target Prediction: Real-time Detection

4.1. USB Camera

Use yolo26n_ncnn_model to predict USB camera footage.

Enter the code folder:

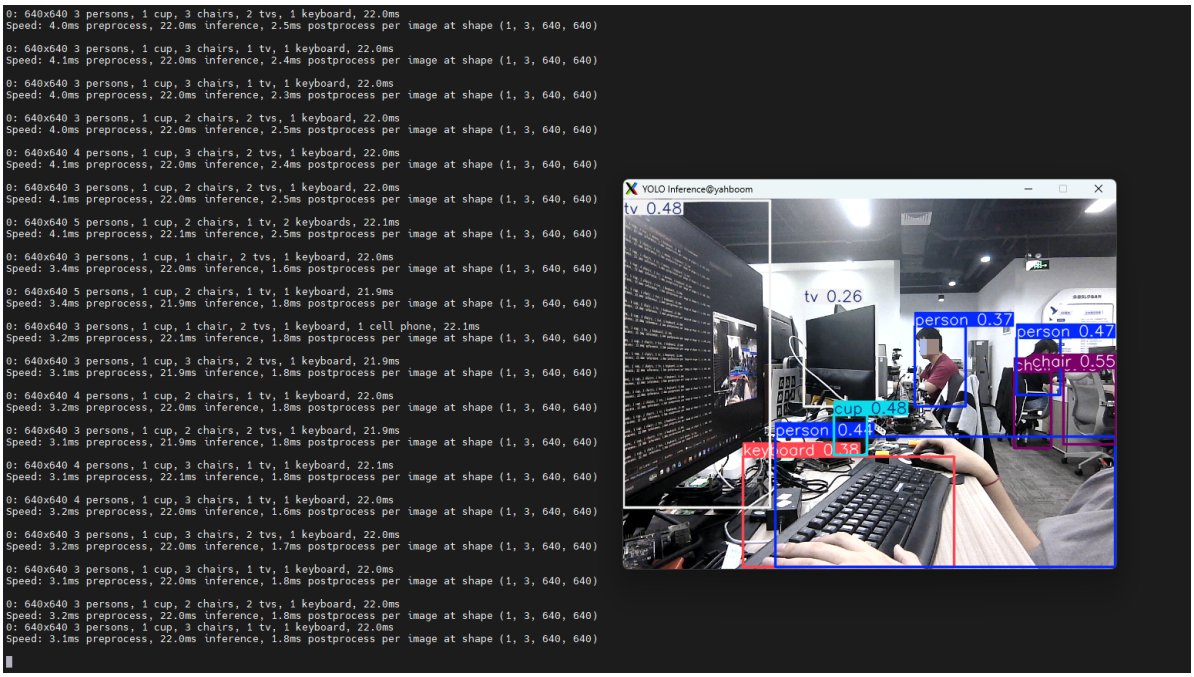
```
cd /home/jetson/ultralytics/yahboom_demo
```

Run the code: Click on the preview screen, press q key to terminate the program!

```
python3 01.detection_camera_usb.py
```

Effect Preview

yolo recognition output video location: /home/jetson/ultralytics/output/



Example code:

```python
import cv2
from ultralytics import YOLO

# Load the YOLO model
model = YOLO("/home/jetson/ultralytics/yolo26n.engine")

# Open the cammera
cap = cv2.VideoCapture(0)
cap.set(6, cv2.VideoWriter.fourcc('M', 'J', 'P', 'G'))
cap.set(cv2.CAP_PROP_FRAME_WIDTH, 640)
cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 480)

# Get the video frame size and frame rate
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = int(cap.get(cv2.CAP_PROP_FPS))

# Define the codec and create a VideoWriter object to output the processed video
output_path = "/home/jetson/ultralytics/output/01.detection_camera_usb.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v')  # You can use 'XVID' or 'mp4v'
depending on your platform
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

# Loop through the video frames
while cap.isOpened():
    # Read a frame from the video
    success, frame = cap.read()

    if success:
        # Run YOLO inference on the frame
        results = model(frame)

        # Visualize the results on the frame
        annotated_frame = results[0].plot()

        # Write the annotated frame to the output video file
        out.write(annotated_frame)


        # Display the annotated frame
        cv2.imshow("YOLO Inference", cv2.resize(annotated_frame, (640, 480)))

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        # Break the loop if the end of the video is reached
        break
# Release the video capture and writer objects, and close the display window
cap.release()
out.release()
cv2.destroyAllWindows()
```

4.2. CSI Camera

> For CSI camera related functions, please be sure to use VNC or directly use a physical display to watch!! Does not support X11 forwarding services like MobaXTerm

Use yolo26n_ncnn_model to predict CSI camera footage.

Enter the code folder:

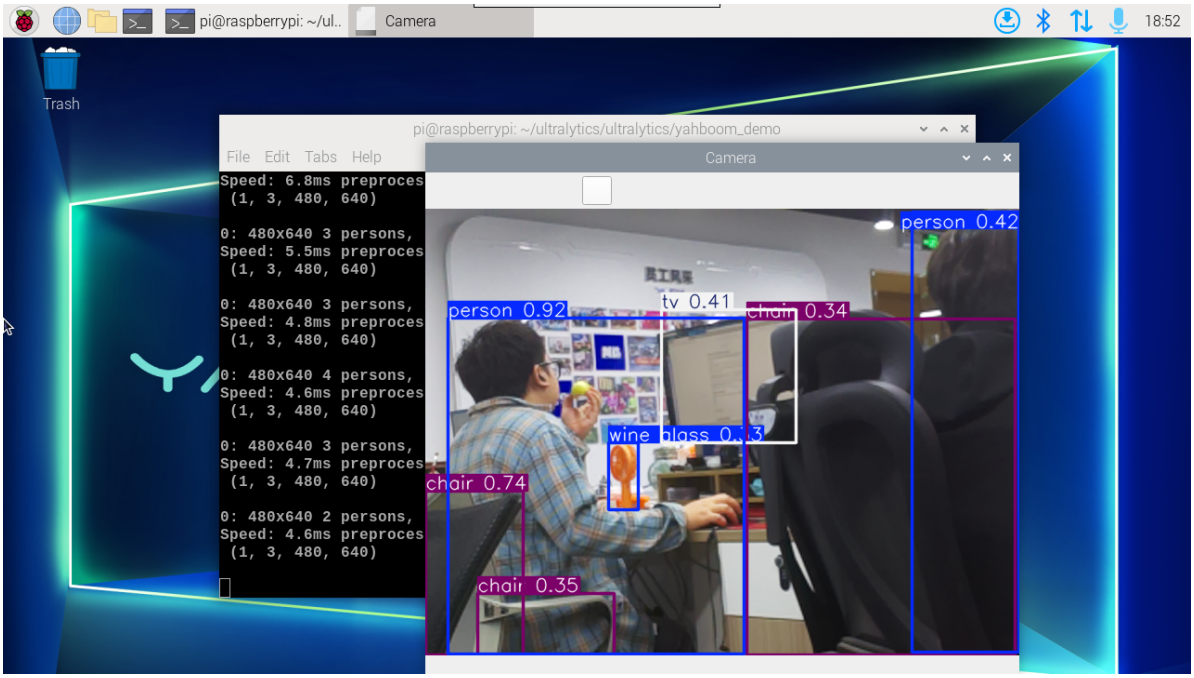
```
cd /home/jetson/ultralytics/yahboom_demo
```

Run the code: Click on the preview screen, press q key to terminate the program!

```
python3 01.detection_camera_csi.py
```

Effect Preview

yolo recognition output video location: /home/jetson/ultralytics/output/



Example code:

```
import cv2
from ultralytics import YOLO
from jetcam.csi_camera import CSICamera

# Load the YOLO model
model = YOLO("/home/jetson/ultralytics/yolo26n.engine")

# Open the camera (CSI Camera)
cap = CSICamera(capture_device=0, width=640, height=480)

# Get the video frame size and frame rate
frame_width = 640
frame_height = 480
fps = 30

# Define the codec and create a VideoWriter object to output the processed video
```

```
output_path = "/home/jetson/ultralytics/output/01.detection_camera_csi.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v')  # You can use 'XVID' or 'mp4v'
depending on your platform
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

# Loop through the video frames
while True:
    # Read a frame from the camera
    frame = cap.read()

    if frame is not None:
        # Run YOLO inference on the frame
        results = model(frame)

        # Visualize the results on the frame
        annotated_frame = results[0].plot()

        # Write the annotated frame to the output video file
        out.write(annotated_frame)

        # Display the annotated frame
        cv2.imshow("YOLO Inference", cv2.resize(annotated_frame, (640, 480)))

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        # Break the loop if no frame is received (camera error or end of stream)
        print("No frame received, breaking the loop.")
        break

# Release the video capture and writer objects, and close the display window
cap.release()
out.release()
cv2.destroyAllWindows()
```

Reference Materials

Object Detection - Ultralytics YOLO Documentation